

59 | OVERVIEW



Powering the Future: Domestic Renewables Versus Carbon Offsets

Abstract

Before the COP26 climate summit, New Zealand raised its pledge to cut greenhouse gas emissions to 50% below 2005 levels by 2030, but planned to buy about two-thirds of those cuts as overseas carbon credits. Climate scientist Kevin Trenberth argued the country should instead cut its own emissions, using rooftop solar, wind, and pumped hydro storage, and remove barriers that discourage homeowners from generating clean power.

Introduction

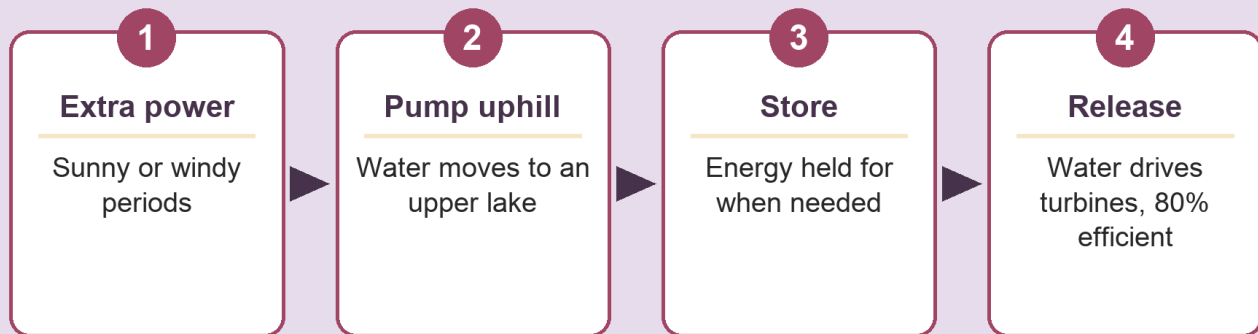
Carbon dioxide has increased 48% since the 1800s, bringing rising temperatures, heavier rains, stronger storms, and more intense droughts, heat waves, and wildfires. A price on carbon, through a tax or trading scheme, can capture the cost of these harms; New Zealand has an Emissions Trading Scheme, though it excludes agriculture. New Zealand's Zero Carbon Act of 2019 requires emissions other than biogenic methane to reach net zero by 2050. Yet the country's plan relied heavily on buying offsets from other countries, which could cost billions, with no system to ensure the cuts are genuine. Trenberth asked: why not generate clean power at home?

Cover: The pumped hydro scheme considered for Lake Onslow would be hundreds of times bigger than existing generation there. Photo: Supplied, via Newsroom.

How Do Scientists Study This?

Trenberth compared New Zealand's solar use with Germany's, which has about 20% less sunshine. He examined how electricity companies pay homeowners for rooftop solar power compared with how much they charge, and contrasted this with net metering used in many countries. He then addressed a key engineering challenge: solar and wind are intermittent. He evaluated storage options, including batteries, hydrogen, and pumped hydro, comparing their efficiency and lifetime, and looked at real examples, such as Switzerland's pumped hydro and a new project at Kidston, Queensland, which repurposes an old gold mine.

How pumped hydro stores clean energy



Compare countries

Measure rooftop solar uptake in New Zealand against sunnier-than-expected Germany.

Follow the money

Compare what companies pay for home solar power with what they charge.

Solve intermittency

Evaluate batteries, hydrogen, and pumped hydro for storing clean energy.

What Did They Discover?

In Germany, 1.3 million homes (3%) have solar, generating 4,900 MW, or 8.2% of the country's power. In New Zealand, about 19,500 homes (1%) generate an estimated 68 MW, and all solar provides just 0.5% of power. A major reason is disincentives: companies typically buy home solar power at \$0.08 per kWh and sell it back at about \$0.28, instead of using net metering. To handle intermittency, wind and solar can be coupled with hydropower by saving water behind dams. Pumped hydro, which pumps water uphill and releases it later, is about 80% efficient and lasts 75 years or more. Switzerland gets 33% of its power capacity from pumped hydro. Trenberth concluded that offsets should be avoided.

Rooftop solar compared

8.2%

of Germany's power from home solar

0.5%

of New Zealand's power from all solar

\$0.08/\$0.28

NZ buy and sell price per kWh for home solar

Sunshine is most abundant when wind and rain are lower, so solar, wind, and hydro naturally complement each other in a national system.

Takeaways

Cut at home

Buying overseas offsets is costly and uncertain; domestic cuts are real.

Reward rooftop solar

Net metering would remove a major barrier to home solar power.

Store the sunshine

Pumped hydro is an efficient, long-lasting way to balance wind and solar.

Key Terms

Carbon offset (credit)

A payment for emissions cuts made elsewhere, used to balance a country's or company's own emissions.

Net metering

A system where extra power from home solar is sent to the grid and credited back at no extra cost.

Pumped hydro storage

Storing energy by pumping water uphill and releasing it through turbines when power is needed.

Emissions Trading Scheme

A market that puts a price on greenhouse gas emissions by requiring companies to hold units for what they emit.

Megawatt (MW)

One million watts, a unit used to measure the output of power stations.

Match-Up: draw a line from each term to its clue

A Carbon offset (credit)

B Net metering

C Pumped hydro storage

D Emissions Trading Scheme

E Megawatt (MW)

1 One million watts, a unit used to measure the output of power...

2 A payment for emissions cuts made elsewhere, used to balance a country's...

3 A system where extra power from home solar is sent to the...

4 Storing energy by pumping water uphill and releasing it through turbines when...

5 A market that puts a price on greenhouse gas emissions by requiring...

Answer key: A-2 · B-3 · C-4 · D-5 · E-1

Activity

Solar Payback Calculator

Students calculate how pricing rules affect whether rooftop solar pays off for a household.

Materials: Calculator; Worksheet with sample household solar output and usage; Graph paper

1. Assume a home sends 1,000 kWh of extra solar power to the grid each year and later uses the same amount back.
2. Calculate the household's cost under New Zealand's system: sold at \$0.08, bought back at \$0.28 per kWh.
3. Calculate the cost under net metering, where the exchange is free.
4. Compare the yearly difference and discuss how it affects the decision to install solar.
5. Write a recommendation to the government comparing domestic solar with buying overseas offsets.

References

Trenberth, K. E. Climate change: NZ should cut its own emissions, not buy overseas offsets. Newsroom.
<https://www.newsroom.co.nz/kevin-trenberth-cut-emissions-dont-buy-overseas-offsets>